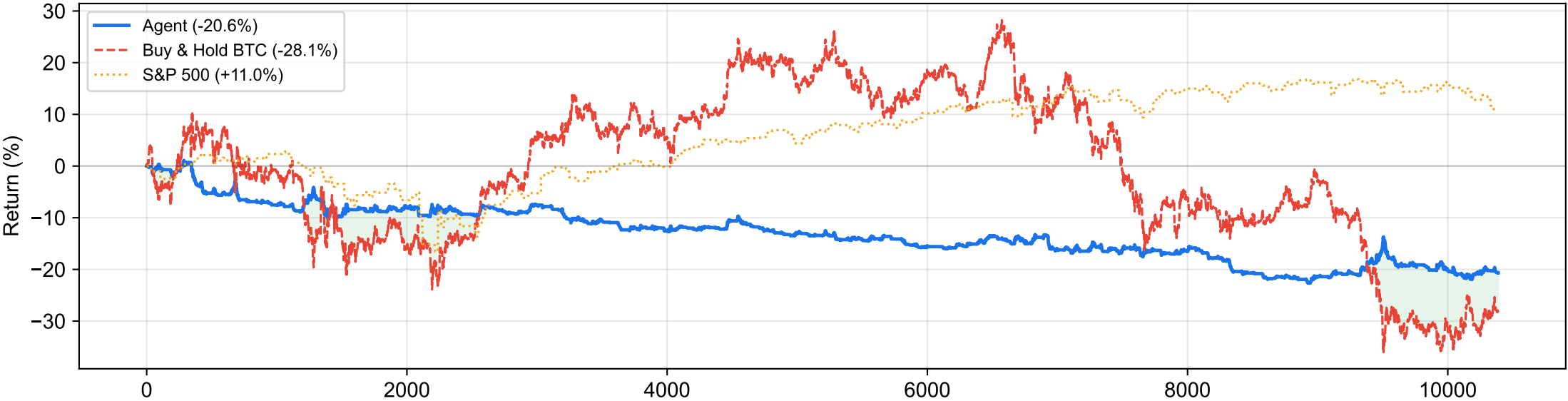
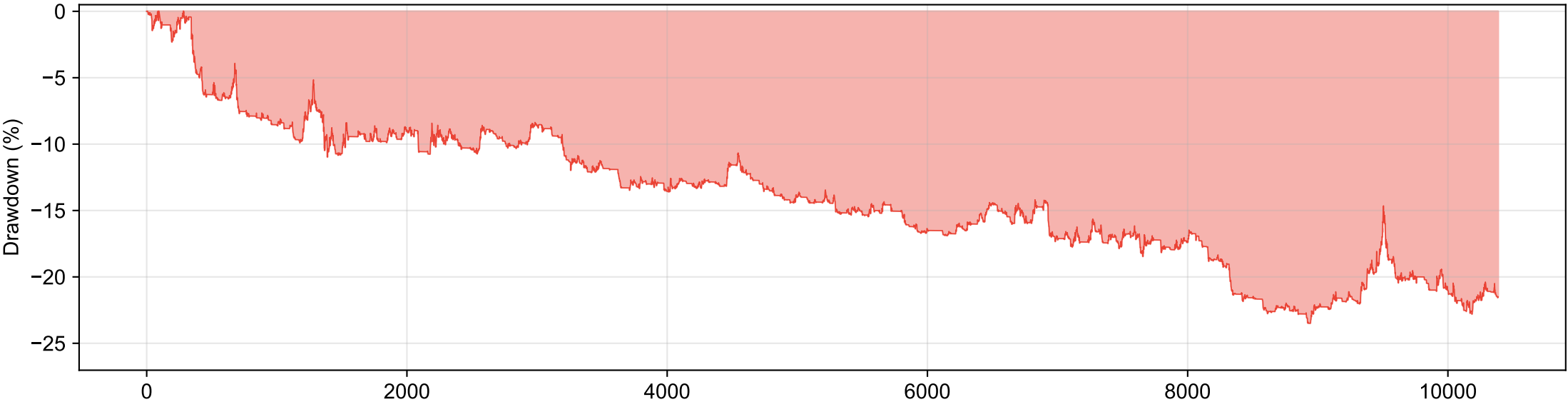


Evaluation Report: CNN+LSTM PPO × SPA (Pure OOS)
Senior AI Project — Out-of-Sample Results

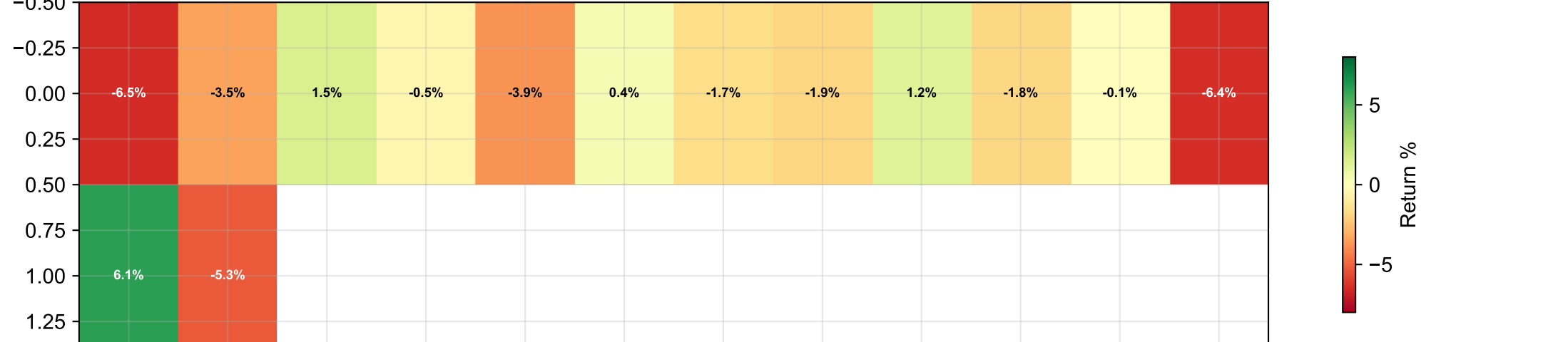
Cumulative Returns: Agent vs Buy & Hold & S&P 500



Drawdown (Underwater Plot)



Monthly Returns Heatmap



Metric		Value
Total Return (Agent)		-20.64%
Total Return (BTC Buy & Hold)		-28.10%
Alpha vs BTC B&H		+7.46%
Total Return (S&P 500)		+10.99%
Performance Summary		
Alpha vs S&P 500		-31.63%
Annualized Return		-17.71%
Annualized Volatility		10.94%
Sharpe Ratio (OOS)		-1.619
Sortino Ratio		-1.680
BENCHMARK SHARPE / SORTINO		
BTC Buy & Hold Sharpe		-0.527
BTC Buy & Hold Sortino		-0.671
S&P 500 Sharpe		0.517
S&P 500 Sortino		0.112
Max Drawdown		-23.50%
Win Rate		36.2%
Profit Factor		0.764
Total Trades		367
Avg Monthly Cashflow		\$337.08